

Lecture/Week 5

Interpolation

Numerical approximation addresses both

- a) Approximating numbers
- b) Approximating functions

The mathematical concept of interpolation is concerned with the problem of approximating a function $f(x)$ defined over a closed interval $[a, b]$.

- The function $f(x)$ to be approximated is usually defined through a set of its values at $n+1$ distinct points lying in the interval $[a, b]$. That is

$$(x_i, f(x_i)) \text{ for } i=0, 1, 2, \dots, n$$

where $x_0 = a$ and $x_n = b$.

Given Data points

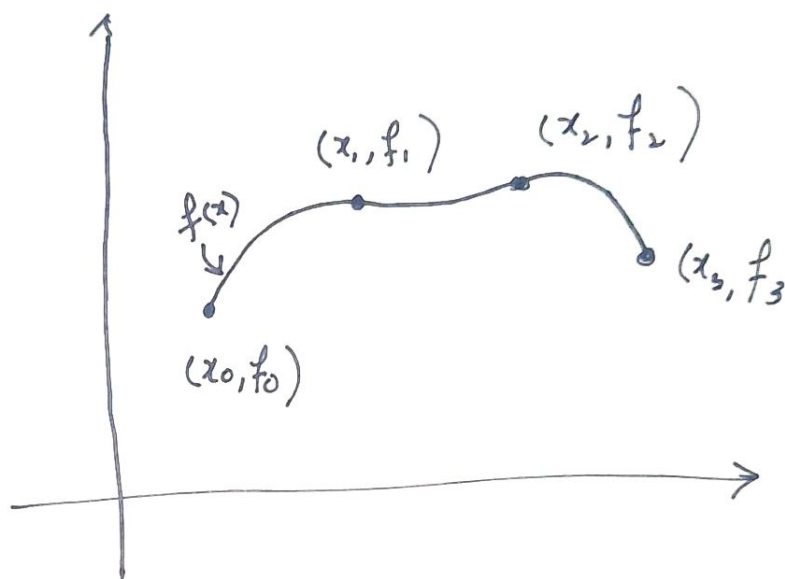
$$(x_0, f(x_0)) (x_1, f(x_1)) (x_2, f(x_2)) \dots (x_n, f(x_n))$$

$$\text{or } (x_0, f_0) (x_1, f_1) (x_2, f_2) \dots (x_n, f_n)$$

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The problem is to determine a function .

$f(x)$ that passes through the distinct points
 $(x_0, f_0) (x_1, f_1) (x_2, f_2) \dots (x_n, f_n)$.



Two types of interpolation

- 1) Polynomial Interpolation
- 2) Spline Interpolation
(Piece-wise continuous function)

Polynomial Interpolation

a) Lagrange Interpolation Method

b) Newton's Divided Difference Interpolation Method

↳ Newton's Forward Difference Interpolation Method

↳ Newton's Backward Difference Interpolation Method

Lagrange Interpolation Method.

Given $(x_0, f_0) (x_1, f_1) (x_2, f_2) \dots (x_n, f_n)$

Problem is to find

$f(x) = P_n(x) \rightarrow$ that is, polynomial of degree 'n'.

This means if we have $n+1$ points the polynomial is of degree 'n'.

Ex:- If we have 10 points, polynomial of degree is 9.

For 100 points, polynomial degree is 99.

" 5 " " " 4.

Formula.

$$P_n(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2 + \dots + L_n(x)f_n$$

where $L_i(x)$ for $i = 0, 1, 2, \dots, n$

are called the Lagrange interpolation coefficients.

$$L_i(x) = \frac{(x-x_0)(x-x_1)(x-x_2)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)(x_i-x_1)(x_i-x_2)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)}$$

$$L_i(x) = \frac{(x-x_0)(x-x_1)(x-x_2)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)(x_i-x_1)(x_i-x_2)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)}$$

For example

$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)\dots(x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)\dots(x_2-x_n)}$$

$$L_5(x) = \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)(x-x_6)(x-x_7)\dots(x-x_n)}{(x_5-x_0)(x_5-x_1)(x_5-x_3)(x_5-x_4)(x_5-x_6)(x_5-x_7)\dots(x_5-x_n)}$$

Example ①

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Given $(1, 1.54)$ $(2, 0.58)$ $(3, 0.01)$ $(4, 0.35)$

<u>x</u>	<u>f</u>
$x_0 = 1$	$f_0 = 1.54$
$x_1 = 2$	$f_1 = 0.58$
$x_2 = 3$	$f_2 = 0.01$
$x_3 = 4$	$f_3 = 0.35$

Question Construct the Lagrange Interpolation polynomial to approximate the value of the function at $x = 2.6$

Solution

Step I ∴ Approximate polynomial function

$f(x) = P_3(x)$ → polynomial of degree 3.

Step II ∴

Find $f(2.6) = P_3(2.6) = \boxed{\dots\dots\dots}$

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Step 1

$$f(x) = P_3(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2 + L_3(x)f_3 \leadsto \textcircled{1}$$

Now

$$\begin{aligned} L_0(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x-2)(x-3)(x-4)}{(1-2)(1-3)(1-4)} \\ &= -\frac{1}{6}(x-2)(x-3)(x-4) \leadsto \textcircled{2} \end{aligned}$$

$$\begin{aligned} L_1(x) &= \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} = \frac{(x-1)(x-3)(x-4)}{(2-1)(2-3)(2-4)} \\ &= \frac{1}{2}(x-1)(x-3)(x-4) \leadsto \textcircled{3} \end{aligned}$$

$$\begin{aligned} L_2(x) &= \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-1)(x-2)(x-4)}{(3-1)(3-2)(3-4)} \\ &= -\frac{1}{2}(x-1)(x-2)(x-4) \leadsto \textcircled{4} \end{aligned}$$

$$\begin{aligned} L_3(x) &= \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)} = \frac{(x-1)(x-2)(x-3)}{(4-1)(4-2)(4-3)} \\ &= \frac{1}{6}(x-1)(x-2)(x-3) \leadsto \textcircled{5} \end{aligned}$$

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using (2) — (5) into (1)

$$P_3(x) = \frac{-1(x-2)(x-3)(x-4)}{6} [1.54] + \frac{1(x-1)(x-3)(x-4)}{2} [0.58] \\ - \frac{1(x-1)(x-2)(x-4)}{2} [0.01] + \frac{1(x-1)(x-2)(x-3)}{6} [0.35]$$

Interpolating
Polynomial.

Interpolation Conditions

$$P_n(x_i) = f_i \quad i = 0, 1, 2, \dots, n.$$

Check for our example

$$P_3(x_i) = f_i \quad i = 0, 1, 2, 3.$$

Let $i = 1$

$$P_3(x_1) = f_1$$

$$P_3(x_1) = P_3(2) = \frac{-1(\cancel{2-2})(2-3)(2-4)}{6} [1.54] + \frac{1(\cancel{2-1})(2-3)(2-4)}{2} [0.58] \\ - \frac{1(\cancel{2-1})(\cancel{2-2})(2-4)}{2} [0.01] + \frac{1(\cancel{2-1})(\cancel{2-2})(2-3)}{6} [0.35] \\ = \frac{1(-1)(-1)(-2)}{2} [0.58] = \underline{\underline{0.58 = f_1}}$$

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Similarly one can check

$$P_3(x_3) = f_3 \quad \text{or} \quad P_3(x_0) = f_0$$

$$P_2(x_2) \text{ or } P_2(x_1) = f_2.$$

Step II ∴

$$f(x) = P_3(x)$$

$$\begin{aligned} f(2.6) = P_3(2.6) &= \frac{-1}{6}(2.6-2)(2.6-3)(2.6-4)[1.54] + \\ &\frac{1}{2}(2.6-1)(2.6-3)(2.6-4)[0.58] - \frac{1}{2}(2.6-1)(2.6-2)(2.6-4)[0.01] \\ &+ \frac{1}{6}(2.6-1)(2.6-2)(2.6-3)[0.35] = \end{aligned}$$

Ans.

Example (2)

x	f
$x_0 \rightarrow 0.1$	$0.91 \rightarrow f_0$
$x_1 \rightarrow 0.2$	$0.70 \rightarrow f_1$
$x_2 \rightarrow 0.3$	$0.43 \rightarrow f_2$
$x_3 \rightarrow 0.5$	$0.54 \rightarrow f_3$

find $f(0.4) = ?$ Solution

$$f(x) = P_3(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2 + L_3(x)f_3$$

$$= \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f_3$$

$$= \frac{(x-0.2)(x-0.3)(x-0.5)}{(0.1-0.2)(0.1-0.3)(0.1-0.5)} (0.91) + \frac{(x-0.1)(x-0.3)(x-0.5)}{(0.2-0.1)(0.2-0.3)(0.2-0.5)} (0.70) + \frac{(x-0.1)(x-0.2)(x-0.5)}{(0.3-0.1)(0.3-0.2)(0.3-0.5)} (0.43) + \frac{(x-0.1)(x-0.2)(x-0.3)}{(0.5-0.1)(0.5-0.2)(0.5-0.3)} (0.54)$$

$$f(0.4) = P_3(0.4) = \text{---} \text{---} \text{---}$$